Use Case Specification: UC-20

- **ID / Code:** UC-20
- **Use Case Name:** Reconcile Bank Payments
- **Description:** The system periodically obtains bank transactions (includes UC-21) and automatically matches them with outstanding invoices using reference number, amount and metadata, updating invoice and account standing (UR-2.4, FR-3.2, FR-3.4). Ambiguous transactions trigger UC-22 through the extension point **ambiguous transaction**.
- **Primary Actor:** None (Automated / Time-triggered process). Supporting actor: Banking API (via UC-21)
- **Scope:** Subscription Management & Invoicing System
- **Level:** User Goal
- **Stakeholders & Interests:**
  - *Company (Product Owner):* Wants automatic, fast payment verification (FR-3.2, NFR-1.3).
  - *Administrator:* Wants to oversee payment reconciliation and have only doubtful cases require manual review (UR-2.4, FR-3.3).
  - *Customer company:* Wants paid invoices to stop generating reminders promptly.
- **Preconditions:** The bank integration or simulated payment environment is configured (FR-3.1, FR-3.4); there are outstanding invoices ("sent", "overdue", "partially reconciled").
- **Postconditions:** Matched invoices are "paid" or "partially reconciled"; ambiguous transactions are queued for review.
- **Success Guarantee:** Each clearly matched transaction updates the invoice status (FR-2.4) and the account standing (FR-2.5.2); administrators are notified of paid invoices (FR-4.2.1).
- **Minimal Guarantee:** No invoice is marked as paid unless the match is unambiguous (FR-3.3); each transaction is processed in isolation (NFR-6.1) and never applied twice (NFR-2.2).
- **Trigger:** Scheduled execution of the reconciliation process (frequency allowing payment status to be reflected within 1 minute – NFR-1.3).
- **Main Success Scenario:**

1. **System:** Initiates the periodic reconciliation process.
2. **System: Includes UC-21** to fetch new bank or simulated payment transactions.
3. **System:** For each transaction not yet processed, searches outstanding invoices by reference number, amount and metadata (FR-3.2).
4. **System:** Finds a single invoice matching exactly.
5. **System:** Links the transaction to the invoice and marks it as "paid".
6. **System:** Updates the customer's account standing (FR-2.5.2), cancels pending reminders for that invoice and notifies administrators (FR-4.2.1).
7. **System:** Records the execution result in the system health dashboard (FR-6.10).

- **Extensions:**
  - 2a. Bank integration error:
    - 2a1. System logs the error, alerts administrators (FR-4.2.2) and ends the execution; it will retry on the next run.
  - 3a. Transaction already processed:

  - 3a1. System skips it to avoid double application (NFR-2.2).
- 4a. Amount lower than the invoice total:
  - 4a1. System marks the invoice as “partially reconciled” and keeps the outstanding balance (FR-2.4).
- 4b. Ambiguous transaction (extension point): no match, several possible invoices, duplicate or delayed transaction:
  - 4b1. System flags the transaction for manual review without marking any invoice as paid (FR-3.3).
  - 4b2. System alerts administrators; UC-22 (Review Unmatched Payment) handles it.